BSc CS IT — PRACTICAL FILE

PRACTICAL / TUTORIAL NO. 8

BUGZILLA AND TESTLINK

Subject: Software Testing

Student Name	RUTVIK VALA
Enrollment Number	2504030400048
Age	20 years
Program	BSc CS IT
Practical Topic	Test case management, test execution reporting, and defect tracking

Includes workbook worksheets and six supplied screenshots for practical reference.

Implementation

Tutorial 8 — Bugzilla & TestLink | Implementation Details	
Parameter	Details
Project Name	Student Login & Registration System
Application/Software	Bugzilla and TestLink
Module Name	Login and Registration
SDLC Model	Waterfall Model (assumed for this academic example)
Programming Language	Not applicable to the tools; sample application is treated as a web application
IDE/Tool Used	Web browser, Bugzilla, TestLink
Input/Test Data	Valid username/password, invalid password, blank fields, invalid email
Student Name	Rutvik Vala
Enrollment No.	2504030400048
Program	BSc CS IT

Test Case Details

Test Case Details — Columns 1 to 8

Test Case Details — Student Login & Registration System							
Test Case ID	Requirement ID	Module Name	Feature Under Test	Test Scenario	Objective	Test Type	Technique
TC-001	REQ-LOGIN-01	Login	Authentication	Valid login	Verify login with correct credentials	Functional / Black Box	Equivalence Partitioning
TC-002	REQ-LOGIN-01	Login	Authentication	Invalid password	Verify login rejects incorrect password	Functional / Black Box	Equivalence Partitioning
TC-003	REQ-LOGIN-02	Login	Form validation	Empty fields	Verify required-field validation	Functional / Black Box	Negative Testing
TC-004	REQ-REG-01	Registration	Form validation	Invalid email	Verify invalid email format is rejected	Functional / Black Box	Equivalence Partitioning

Test Case Details — Columns 9 to 20

Priority	Severity	Preconditions	Postconditions	Test Environment	Browser/OS	Test Data	Expected Result	Actual Result	Status	Executed By	Execution Date
High	N/A	User account exists	User session starts	Web application	Chrome / Windows	student01 / ValidPass123	Dashboard opens successfully	Dashboard opens successfully	Pass	Student QA	28-10-2026
High	Major	User account exists	Defect logged if access granted	Web application	Chrome / Windows	student01 / WrongPass	Access denied and error message appears	Login accepted wrong password in sample scenario	Fail	Student QA	28-10-2026
Medium	N/A	None	No login session created	Web application	Chrome / Windows	Username blank; password blank	Required-field messages appear	Required-field messages appear	Pass	Student QA	28-10-2026
Medium	N/A	Registration page available	Registration is not submitted	Web application	Chrome / Windows	student@	Validation message appears	Validation message appears	Pass	Student QA	28-10-2026

Test Execution

Test Case Execution						
Step No.	Test Case ID	Test Steps	Input	Expected Result	Actual Result	Status
1	TC-001	Open login page	Test application URL	Login form is displayed	Login form displayed	Pass
2	TC-001	Enter valid username and password	student01 / ValidPass123	Credentials accepted	Credentials accepted	Pass
3	TC-001	Click Login	Click Login button	Dashboard opens	Dashboard opens	Pass
1	TC-002	Open login page	Test application URL	Login form is displayed	Login form displayed	Pass
2	TC-002	Enter valid username and incorrect password	student01 / WrongPass	Access denied	Login incorrectly accepted in sample scenario	Fail
3	TC-002	Click Login	Click Login button	Error message appears	Dashboard opens in sample scenario	Fail
1	TC-003	Leave username and password blank	Blank / Blank	Required-field validation appears	Required-field validation appears	Pass
1	TC-004	Enter invalid email on registration page	student@	Invalid email rejected	Validation message appears	Pass
1	BUG-001	Record defect in Bugzilla	Summary and reproduction steps	Bug record is created	Example ID: BUG-001	Pass

Bugzilla Defect

Bugzilla Defect Record	
Field	Sample Value
Bug ID	BUG-001 (example only)
Product	Student Login & Registration System
Component	Login Module
Summary	Login accepts an incorrect password
Description	The login page allows a user to log in using a valid username and an incorrect password.
Steps to Reproduce	1. Open login page. 2. Enter valid username. 3. Enter incorrect password. 4. Click Login.
Expected Result	System should deny access and display an invalid-credentials message.
Actual Result	Login is accepted in this sample defect scenario.
Severity	Major
Priority	P2
Status	NEW
Resolution	UNRESOLVED

Testing Activities

Testing Activities Performed				
Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable / Artifact
1	Requirement Analysis	Requirement Analysis	Identify login and registration requirements	Requirement notes
2	Design	Test Planning	Prepare scope, approach, and resources	Test plan
3	Design	Test Case Development	Write test cases and test data	Test case suite
4	Development	Test Environment Setup	Prepare browser and test application	Environment details
5	Testing	Test Execution	Execute tests and log defect	Execution record and BUG-001

Execution Report

Test Execution Report						
Test Case ID	Module Name	Test Scenario	Expected Result	Actual Result	Status	Remarks
TC-001	Login	Valid login	Dashboard opens	Dashboard opens	Pass	Working as expected
TC-002	Login	Invalid password	Access denied	Incorrect login accepted in sample	Fail	Log BUG-001
TC-003	Login	Empty fields	Validation appears	Validation appears	Pass	Working as expected
TC-004	Registration	Invalid email	Invalid email rejected	Validation appears	Pass	Working as expected

Summary Report

Test Summary Report	
Metric	Value
Total Test Cases Designed	
Total Test Cases Executed	
Total Passed Test Cases	
Total Failed Test Cases	
Total Blocked Test Cases	0
Total Not Executed Test Cases	0
Pass Percentage	
Fail Percentage	
Overall Test Status	
Tester Name	Student QA
Execution Date	28-10-2026

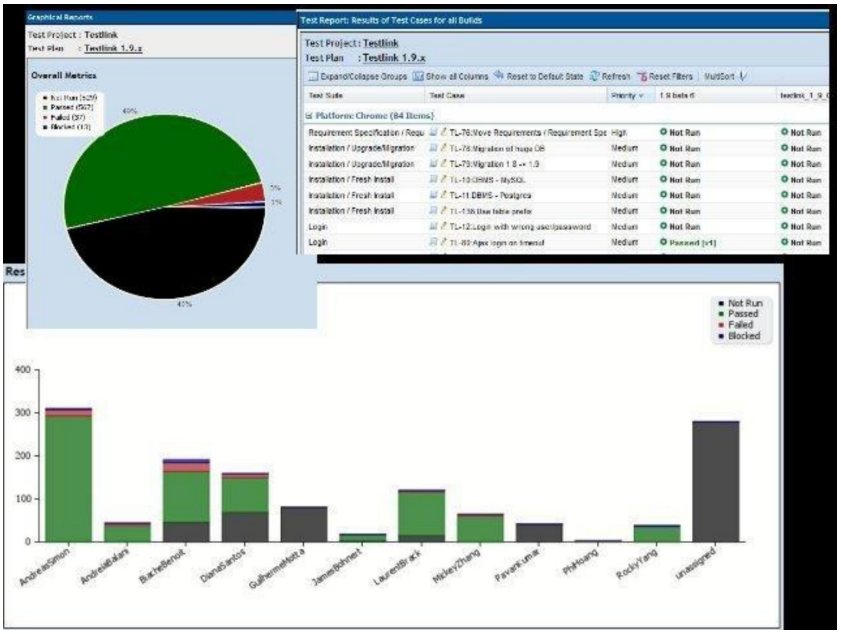
MCQ Answers

Tutorial 8 — MCQ Answers		
Q. No.	Correct Answer	Explanation
1	B) Defect Tracking and Management	Bugzilla is primarily used to report and track software defects.
2	B) Test Case Management	TestLink is used to organize and manage test cases and execution.
3	A) New	A newly reported defect commonly starts with the New status.
4	C) Test Summary Report	It provides an overall view of testing statistics and results.
5	C) Improve Testing Efficiency and Traceability	Testing tools help organize tests and track defects and results.

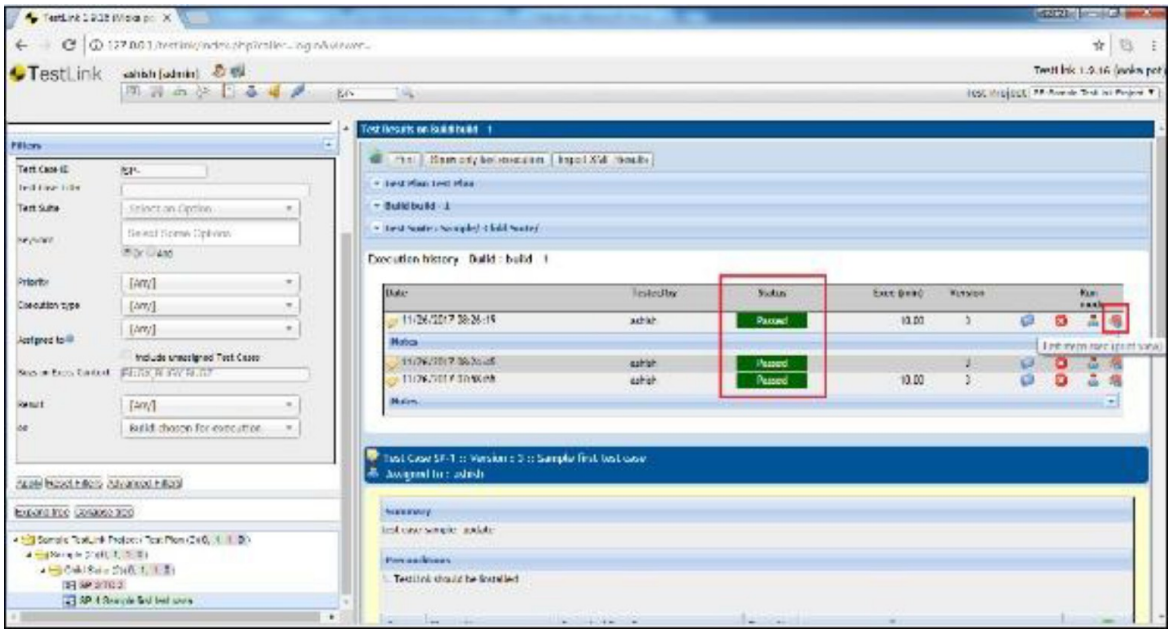
Viva Questions

Viva Questions — Quick Preparation	
Question	Answer
What is Bugzilla?	A defect tracking tool used to report, assign, and monitor software bugs.
What is TestLink?	A test management tool used to create test plans, test suites, test cases, and execution reports.
What is a test case?	A set of steps, inputs, and expected results used to verify a feature.
What is severity?	The impact of a defect on the application.
What is priority?	How urgently a defect should be fixed.
What is a test execution report?	A report showing each test case result, such as Pass or Fail.
What is traceability?	The ability to link requirements to test cases and defects.

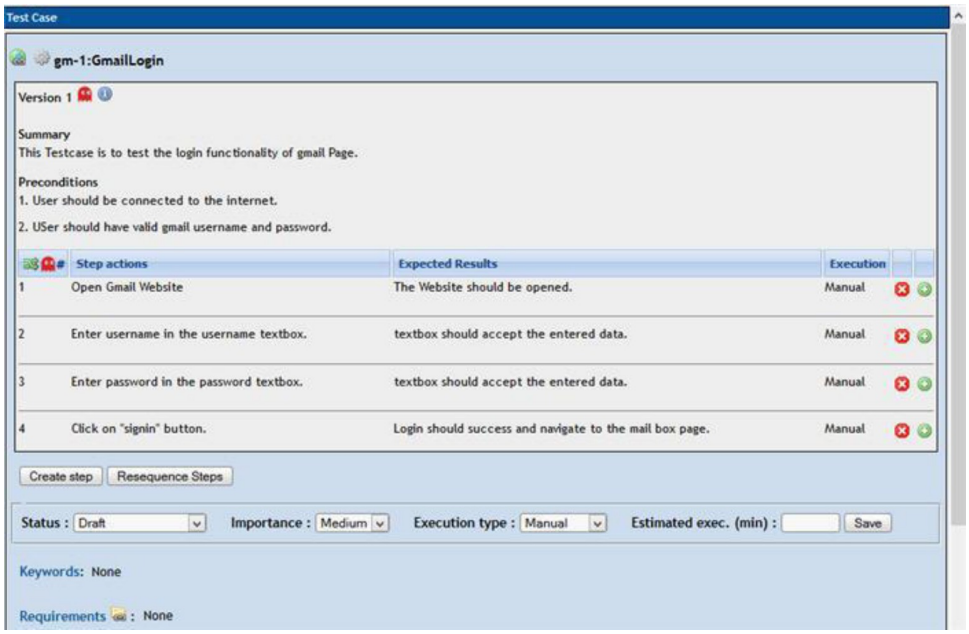
Screenshot 1 — TestLink Graphical Report and Overall Metrics



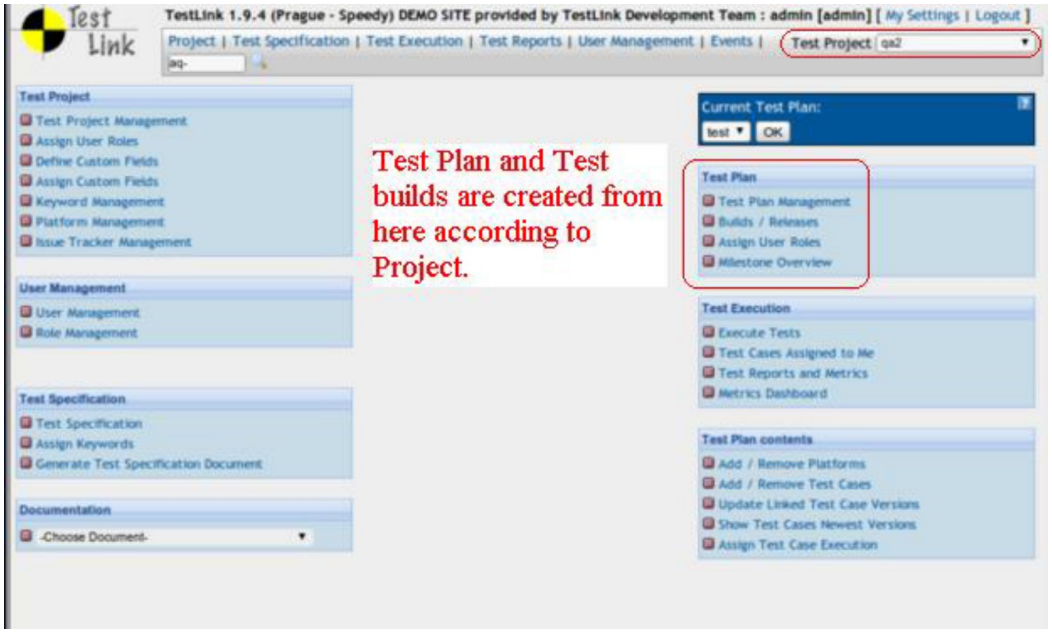
Screenshot 2 — TestLink Test Execution Details



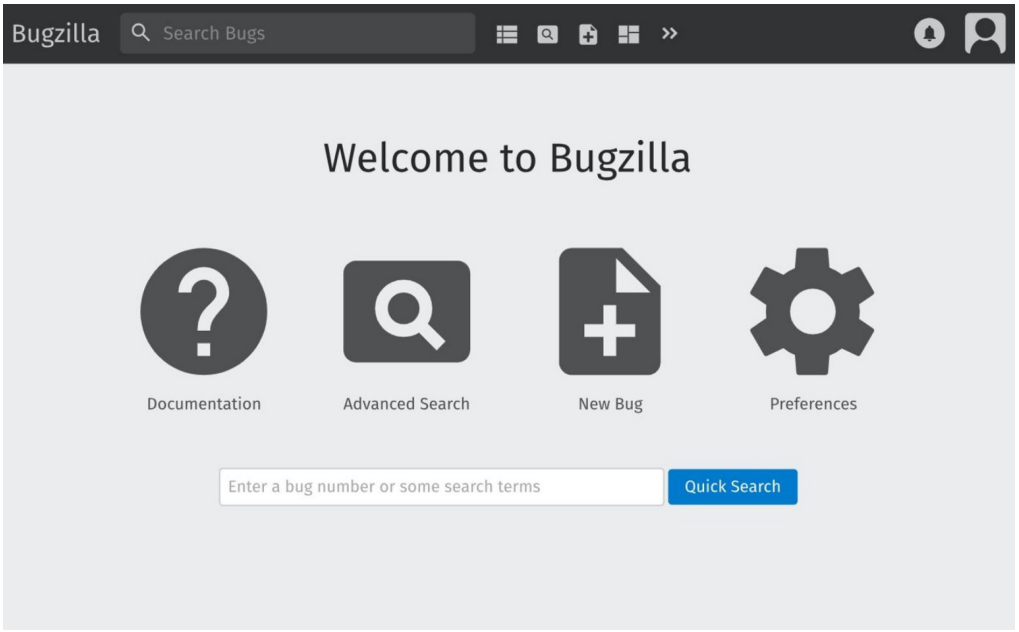
Screenshot 3 — Gmail Login Test Case Specification



Screenshot 4 — TestLink Test Plan and Project Overview



Screenshot 5 — Bugzilla Home Page



Screenshot 6 — Bugzilla New Defect Form

